Introduction to Derivatives

Assignment 2

1) It is now June. A company knows that it will sell 5,000 barrels of crude oil in September. It uses the October CME Group futures contract to hedge the price it will receive. Each contract is on 1,000 barrels of ''light sweet crude.'' What position should it take? What price risks is it still exposed to after taking the position? 2 pts

2) A cattle farmer expects to have 120,000 pounds of live cattle to sell in three months. The live cattle futures contract traded by the CME Group is for the delivery of 40,000 pounds of cattle. How can the farmer use the contract for hedging? From the farmer's viewpoint, what are the pros and cons of hedging? 2 pts

3) A 10-year 8% coupon bond currently sells for $90. A 10-year 4% coupon bond currently sells for $80. The bond coupons are paid every 6 months. What is the 10-year zero rate? (*Hint*: Consider taking a long position in two of the 4% coupon bonds and a short position in one of the 8% coupon bonds.) Assume continuous compounding in all cases. 3 pts

4) The cash prices of 6-month and 1-year Treasury bills are 94.0 and 89.0. A 1.5-year Treasury bond that will pay coupons of $4 every 6 months currently sells for $94.84. A 2year Treasury bond that will pay coupons of $5 every 6 months currently sells for $97.12. Calculate the 6-month, 1-year, 1.5-year, and 2-year continuously compounded Treasury zero rates. It is given to you that 6m Treasury bill yields are based on semi-annual compounding and 1-year Treasury bills are based on annual compounding. 3 pts

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